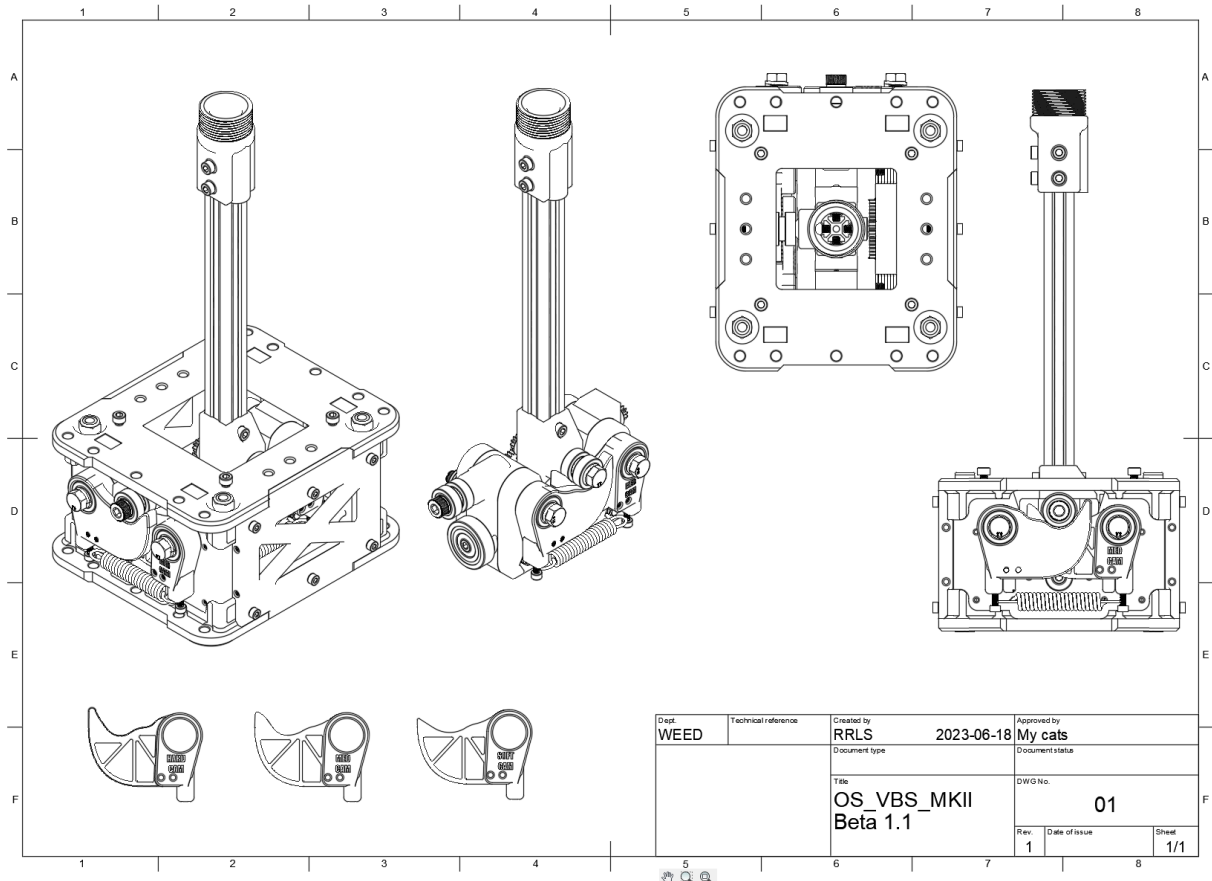




Open Source - Very Big Stick MKII

MECHANICAL BUILD GUIDE

REVISED Beta v1.4.0

2025-04-09



Support this project at <https://www.patreon.com/RightRudderLeftStick>

Open Source - Very Big Stick MKII.....	1
Mechanical Build Guide Video.....	3
Bill of Materials.....	4
STL Part Reference.....	6
PitchFrame_TopPlate.stl (revised 1.4).....	6
PitchFrame_SidePlate.stl.....	6
TM_M36_2020_Adapter.stl.....	6
Cam_rollerSpacer.stl.....	7
Cam_bearing_washer.stl.....	7
MainSleeve.stl (revised 1.4).....	7
PitchFrame_PillowBlock.stl (revised 1.4).....	8
RollFrame_PitchPin_Gear.stl (revised 1.4).....	8
RollFrame_PitchPin_Camside.stl.....	8
RollFrame_GearSide.stl (revised 1.4).....	9
RollFrame_CamSide.stl (revised 1.4).....	9
M12_WasherSpacer.stl.....	9
AS5600_Sensor_Mount_vr3.stl.....	9
CAM Profile Options.....	10
SOFT_CAM.stl.....	11
MED_CAM.stl.....	11
HARD_CAM.stl.....	11



Mechanical Build Guide Video

Instructions on how to assemble the OS_VBS MKII gimbal is available on rightrudder-leftstick's youtube channel. Please watch and follow carefully on how to assemble these components.



You can watch the video at this URL:

<https://www.youtube.com/watch?v=IdYbF5dOYag>



Support this project at <https://www.patreon.com/RightRudderLeftStick>

Bill of Materials

This is a live link to a spreadsheet containing the bill of materials for hardware and parts to be printed.

Note that as this time the Electrical section is just a placeholder

Part name	Quantity	Thicker walls required?	grams each	Total grams	Optional?
PitchFrame_TopPlate	2	n	134	268	
PitchFrame_SidePlate	2	n	47	94	
TM_M36_2020_Adapter	1	y	31	31	Y
Cam_rollerSpacer	2	n	1	2	
Cam_bearing_washer	4	n	1	4	
MainSleeve	1	y	54	54	
PitchFrame_PillowBlock	2	y	95	190	
RollFrame_PitchPin_Gear	1	y	43	43	
RollFrame_PitchPin_Camside	1	y	46	46	
RollFrame_GearSide	1	y	49	49	
RollFrame_CamSide	1	y	49	49	
M12_WasherSpacer	2	n	2	4	Y
SOFT_CAM	4	y	23	92	
TOTAL Filament PETG				926	grams
HARDWARE		Notes			
1KG PETG 1.75mm filament	1	PLA+ untested			
2020 v slot aluminum rod - Cut to length	1	Length depends on your choices			
608 bearings	12				
6003 bearings	4				
18x24.7x1 Teflon Washers	4	https://www.aliexpress.com/item/4000935512060.html			Y
M3x8mm	4	Do not use countersunk heads			
M4x20mm Socket head	20				
M5 vslot drop in nuts	11				
M5x8mm Socket Head	10	6 less if TM adapter is not used			
M5x16mm Socket head	2	Put into the middle of the camside 6003 bearings for extra strength			Y
M5x50mm	2	For the counterweights			Y
M8 washers	4				
M8 Spring Lock washer	6				



M8 Nylon Lock nuts	8		
M8x35mm HEX Head	3	Must be hex head for clearance	
M8x50mm socket head	3	Two of them must be Socket Headed	
M8x80mm HEX or Socket Head	2		
M8x100mm HEX head	4	HEX head only	
M12 Washers	20	For counterweights	Y
FRT-C2-301-G1 Rotary damper	2	Make sure to get the kind that dampens both directions!	Y
Extension spring with loop ends. ~60mm unloaded	2	I used this mcmaster part in the design (3630n293), untested! You'll want to test many springs	
Extension spring with loop ends. various lengths	2	This is the second set of side springs, try several to tweak	Y
ELECTRICAL (WIP)		Notes	
5pin mini din connector (male) for TM adapter	1	https://www.aliexpress.com/item/32859748866.html	y
TLE5010 sensor kit with magnet and PCB	2	https://www.aliexpress.com/item/4000549883214.html	
Assorted 22 AWG wire in various colours	1	get at least a meter worth in several colours.	
STM32F103C8 microcontroller	1	Be careful to get the exact right version that supports Freejoy, beware of counterfeits	
Solder breadboard	1	It will be needed to create a "breakout" board for the TLE5010 and Shift register sensors	
AS5600 white breakout boards	2	This is an alternate to the TLE5010 kit.	

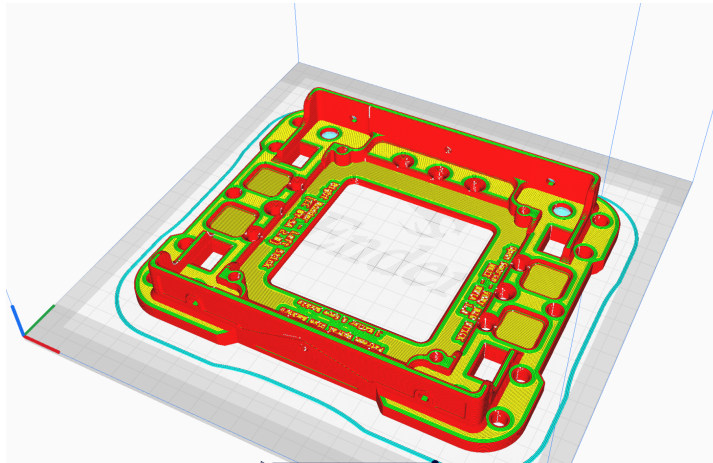


STL Part Reference

Here is information on each stl file included with the build and a reference image showing the correct print orientation.

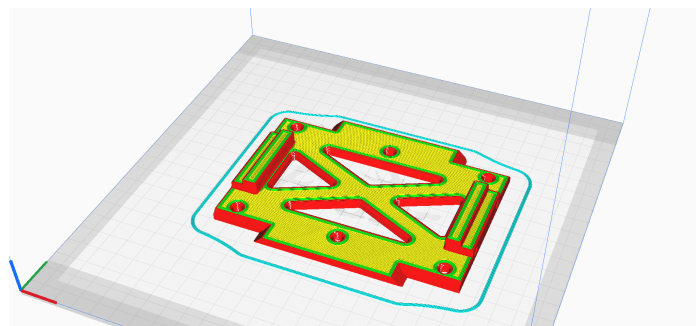
PitchFrame_TopPlate.stl (*revised 1.4*)

Quantity: 2



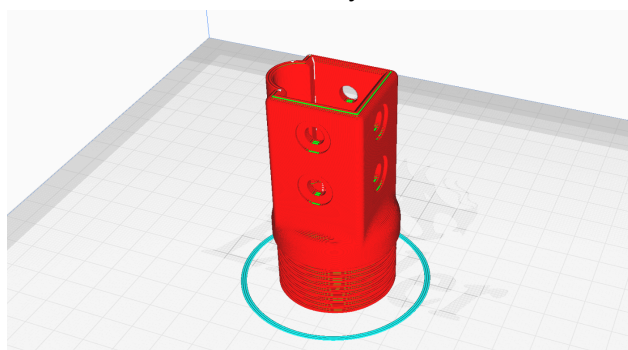
PitchFrame_SidePlate.stl

Quantity: 2



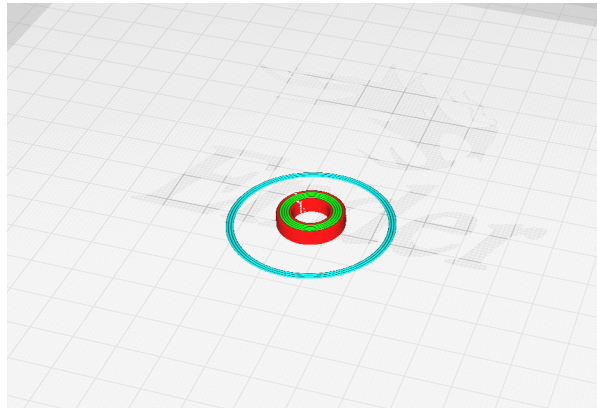
TM_M36_2020_Adapter.stl

Quantity: 1



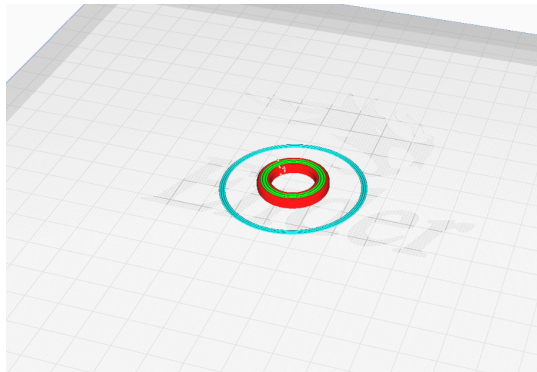
Cam_rollerSpacer.stl

Quantity: 2



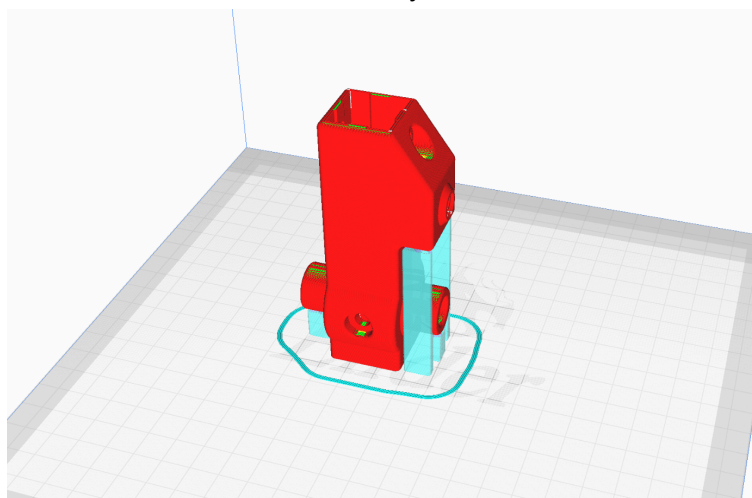
Cam_bearing_washer.stl

Quantity: 4



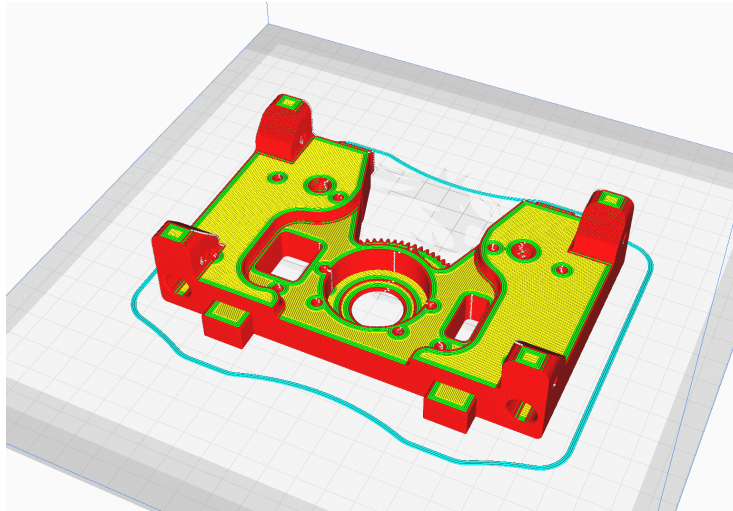
MainSleeve.stl (revised 1.4)

Quantity: 1



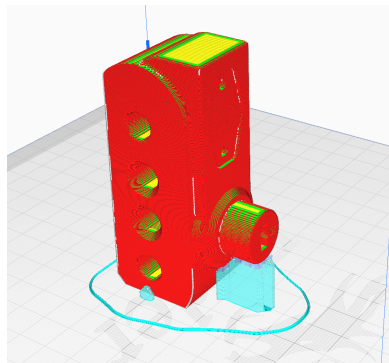
PitchFrame_PillowBlock.stl (*revised 1.4*)

Quantity: 2



RollFrame_PitchPin_Gear.stl (*revised 1.4*)

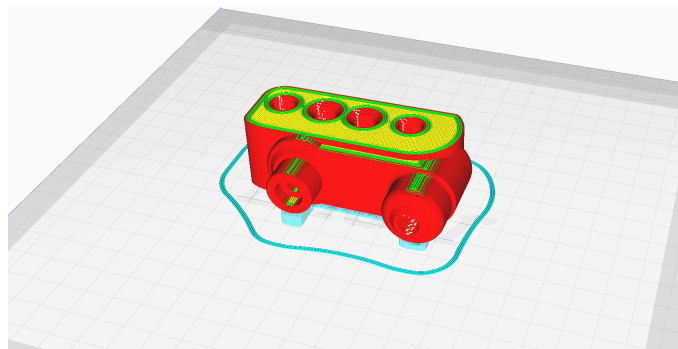
Quantity: 1



*** Note, this has been revised to print vertically for the magnet slot * Beta 1.4.0**

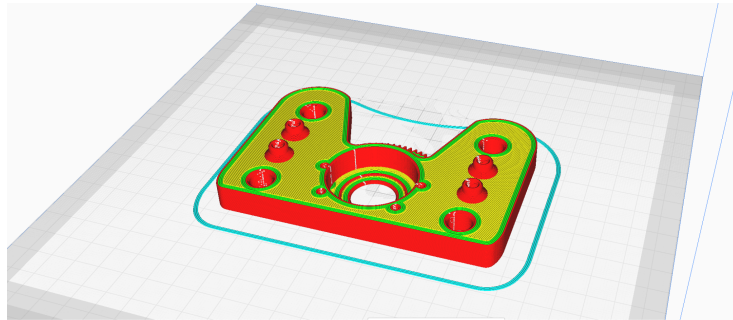
RollFrame_PitchPin_Camside.stl

Quantity: 1



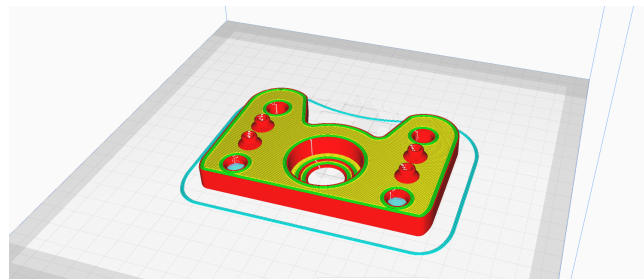
RollFrame_GearSide.stl (revised 1.4)

Quantity: 1



RollFrame_CamSide.stl (revised 1.4)

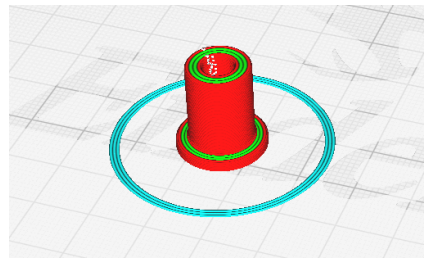
Quantity: 1



M12_WasherSpacer.stl

Quantity: 1

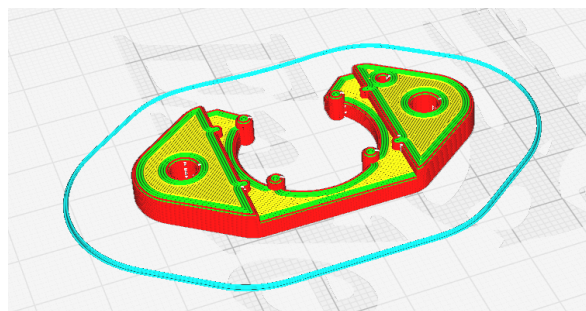
Optional



AS5600_Sensor_Mount_vr3.stl

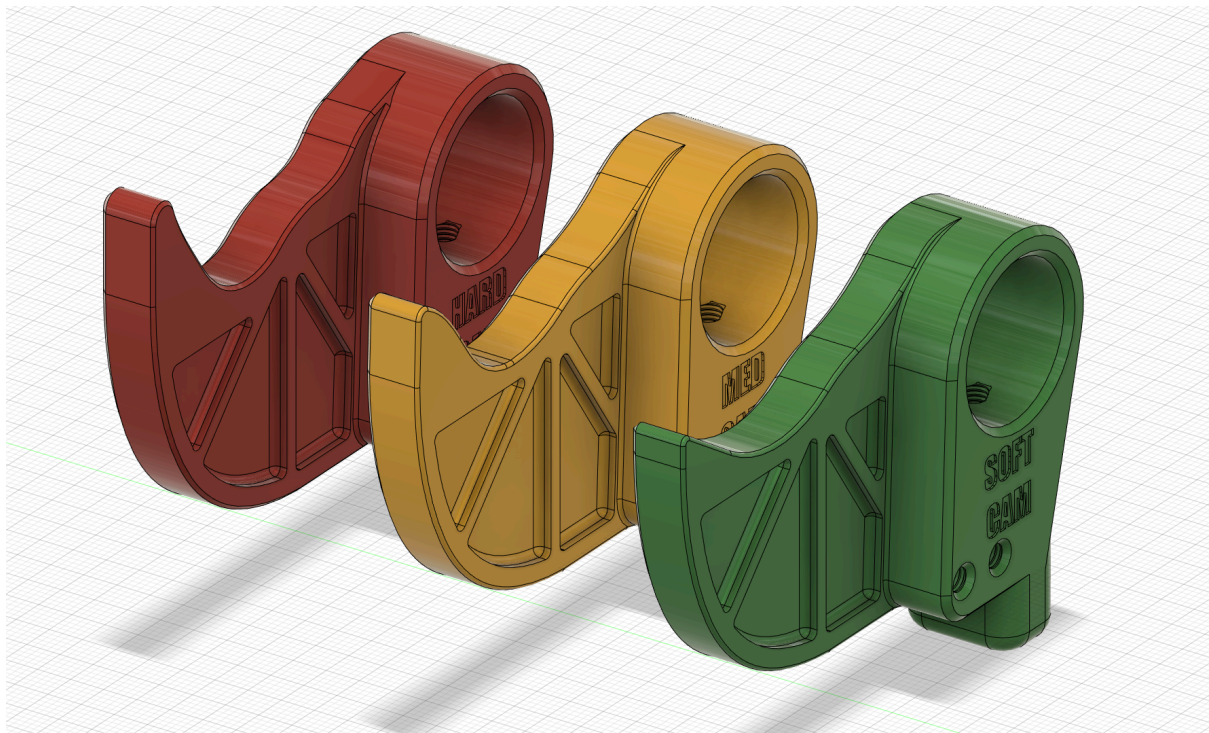
Quantity: 2

Optional



CAM Profile Options

The design currently comes with three different options of CAM profiles, They can be mixed and matched on different axis to produce different force feelings, such as a stiffer pitch but a lighter roll motion. They are scaled from Soft to Hard.

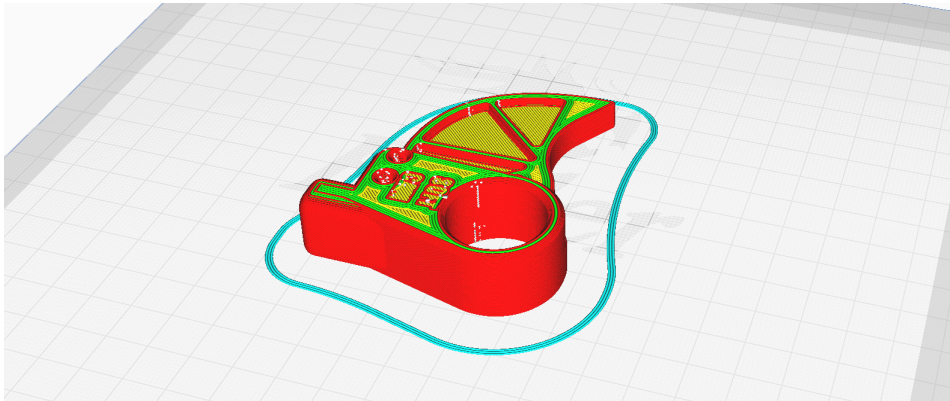


They also feature mounting for a bottom extension spring as well as 4 M4 holes on the side for mounting a secondary side spring. This can either augment the force of the primary spring or allow the use of a very light spring for low spring force applications like helicopter cyclics.



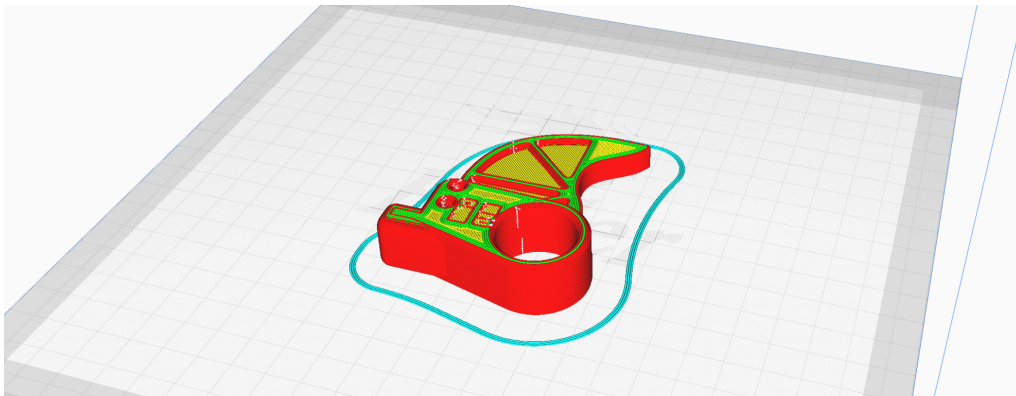
SOFT_CAM.stl

Quantity: 4



MED_CAM.stl

Quantity: 4



HARD_CAM.stl

Quantity: 4

